

CSS Animation & Motion — Interview Cheatsheet

For GFE / frontend-coding loops (OpenAI · Shopify · Airbnb · Figma). Motion is a **polish** layer, not the point — ship correct logic + states + a11y first, then add this as a named step. Everything below with ★ is write-from-memory.

FLUENT ★ (no lookup)
transition, transform (translate/scale/rotate) + why, @keyframes+animation, opacity, easing basics, prefers-reduced-motion, the height-auto & exit-anim traps.

RECOGNIZE (have a plan)
Exit/unmount timing, @starting-style+allow-discrete, FLIP for reorder, will-change hygiene, Web Animations API, scroll-behavior.

SKIP (these loops)
3D/perspective, spring-physics internals, Framer/GSAP internals, canvas/SVG/Lottie animation, staggered choreography, motion-path.

The pixel pipeline — the *why* ★

This is the mechanism answer that separates you. Say it out loud when you pick a property.
Every frame runs some subset of: **Style → Layout → Paint → Composite**. The property you animate decides how much of the pipeline re-runs at 60fps (every ~16ms).

Stage	Triggered by (examples)	Cost
Layout (reflow)	width height top left margin padding	worst
Paint	color background box-shadow border-radius	bad
Composite	transform opacity filter	cheap (GPU)

Rule: animate transform and opacity. They run on the compositor thread — off the main thread, so they stay smooth even while JS is busy. Anything else, justify it.

transition — the workhorse ★

```
/* property | duration | easing | delay */
transition: transform .3s ease-out;
transition: opacity .2s, transform .3s ease .1s; /* multiple */
```

Fires when a property **changes value** (hover, class toggle, state). Needs two interpolatable endpoints.

Avoid transition: all — animates props you didn't mean to (layout jank + surprise transitions). List properties explicitly.

Not animatable the naive way: height: auto, display, and discrete props. See traps below for the fixes.

transform functions ★

translateX/Y() translate()	move (no layout)
scaleX/Y() scale()	resize — your progress-bar tool
rotate()	spin
skewX/Y()	rarely needed

transform-origin sets the anchor (default center). A fill bar using scaleX **must** set transform-origin: left or it grows from the middle.

Order matters: translate() rotate() ≠ rotate() translate().
Modern: individual translate/rotate/scale props animate independently without clobbering each other.

@keyframes + animation ★

Use when there's **no state change to trigger a transition** — looping or multi-step: spinners, skeletons, indeterminate bars, blinking caret.

```
@keyframes spin { to { transform: rotate(360deg); } }
.spinner {
  /* name dur easing delay count direction fill */
  animation: spin 1s linear infinite;
}
```

- iteration-count: infinite · direction: alternate (ping-pong)
- fill-mode: forwards holds the last frame; both also applies frame 0 during delay
- animation-play-state: paused to freeze
- Steps: 0% / 50% / 100% or from / to

Easing / timing functions

ease	default; gentle both ends
linear	spinners, mechanical loops
ease-out	enters (fast→slow, feels snappy)
ease-in	exits (slow→fast, gets out of the way)
cubic-bezier(...)	custom curves; overshoot with y>1
steps(n)	sprite sheets, typewriter, blink

One-liner to say: "ease-out on the way in, ease-in on the way out." It signals taste for almost no words.

The traps — the differentiators ★

1 · height: auto can't transition (accordion)

Modern clean fix — animate a grid track:

```
.panel { display: grid; grid-template-rows: 0fr;
  transition: grid-template-rows .3s ease; }
.panel.open { grid-template-rows: 1fr; }
.panel > .inner { overflow: hidden; } /* required */
```

Fallbacks: `max-height` hack (pick a value $\geq$ real height; easing feels off) · or JS-measure `scrollHeight` and set an explicit px height.

2 · Exit animations (unmount timing)

The classic React miss: you set state $\rightarrow$ element unmounts *instantly* $\rightarrow$ the exit transition never plays because the node is already gone.

Keep it mounted until the transition ends, *then* remove:

```
// on close: play out, unmount on transitionend
<div className={closing ? "toast out" : "toast"}
  onTransitionEnd={() => closing && onDone()} />
```

Real projects reach for `AnimatePresence` (Framer) or `react-transition-group` — name-drop, but know the manual pattern.

3 · display / discrete (modern answer)

```
dialog { opacity: 1;
  transition: opacity .3s, display .3s allow-discrete; }
@starting-style { dialog[open] { opacity: 0; } }
```

Pre-2024 fallback: toggle `visibility` + `opacity` instead of `display`.

Performance & hygiene

- **will-change:** `transform` promotes to its own layer — use **sparingly** and remove after; each layer costs memory.
- Old GPU nudge: `transform: translateZ(0) / translate3d(0,0,0)`.
- Don't animate `box-shadow` (paint-heavy) — animate `opacity` of a pseudo-element holding the shadow.
- JS-driven motion $\rightarrow$ `requestAnimationFrame`, never `setInterval`.
- Debounce animations to **transform/opacity** before blaming the framework.

prefers-reduced-motion ★

Ship this unprompted — it's the highest-signal, lowest-effort a11y move in a motion component.

```
@media (prefers-reduced-motion: reduce) {
  *, ::before, ::after {
    animation-duration: .01ms !important;
    animation-iteration-count: 1 !important;
    transition-duration: .01ms !important;
    scroll-behavior: auto !important;
  }
}
```

Narrate it: "vestibular users get essential motion killed; the component still works."
Definition-of-done item.

Component recipes — progress bar ★

Determinate — two options

```
/* A. width — simplest, but triggers layout */
.fill { width: var(--p); transition: width .3s ease; }

/* B. scaleX — composite-only, the perf answer */
.fill { transform-origin: left;
  transform: scaleX(var(--p)); /* 0..1 */
  transition: transform .3s ease-out; }
```

scaleX caveat: it distorts children (text/icons stretch). Use it for a bare fill; use `width` if the bar has content. Mention this tradeoff — it's the whole point of the question.

A11y: `role="progressbar"` + `aria-valuenow/min/max`.

Indeterminate

```
@keyframes slide { to { transform: translateX(400%); } }
.blip { animation: slide 1.2s ease-in-out infinite; }
```

Recipes — carousel ★

```
.track { display: flex;
  transform: translateX(calc(var(--i) * -100%));
  transition: transform .4s ease; }
```

Infinite loop gotcha: clone first/last slides. When you jump from a clone back to the real slide, **disable the transition for that one frame** or you see it rewind:

```
track.style.transition = "none";
track.style.transform = realOffset;
track.offsetHeight; // force reflow to commit
track.style.transition = ""; // re-enable
```

CSS-only alt for simple cases: `scroll-snap-type: x mandatory` + `scroll-snap-align: start`.

Recipes — modal / drawer / toast

```
.modal { opacity: 0; transform: translateY(8px);
  transition: opacity .2s, transform .2s ease-out; }
.modal.open { opacity: 1; transform: translateY(0); }
.backdrop { opacity: 0; transition: opacity .2s; }
```

- Enter is trivial; the **exit is the interview** — defer unmount (trap #2).
- Drawer = swap `translateY` for `translateX(-100%)<0`.
- Pair with focus-trap + `role="dialog"` + `aria-modal`.

Recipes — spinner · skeleton · tabs

Spinner

```
@keyframes spin { to { transform: rotate(360deg); } }
.spinner { animation: spin .8s linear infinite; }
```

Skeleton shimmer

```
.skel { background: linear-gradient(90deg,
  #eee 25%, #f5f5f5 50%, #eee 75%);
  background-size: 200% 100%;
  animation: shimmer 1.4s linear infinite; }
@keyframes shimmer { to { background-position-x: -200%; } }
```

Tab indicator

Measure active tab's `offsetLeft/offsetWidth`, then slide one bar:

```
.ink { transform: translateX(var(--x));
  width: var(--w); transition: transform .25s, width .25s; }
```

OpenAI streaming chat — motion bits ★

Your double-starred build. Motion here is minimal and mostly about not janking under high-frequency updates.

Blinking caret

```
@keyframes blink { 50% { opacity: 0; } }
.caret { animation: blink 1s step-end infinite; }
```

Typing dots (staggered)

```
.dot { animation: bob 1.2s ease-in-out infinite; }
.dot:nth-child(2){ animation-delay:.15s; }
.dot:nth-child(3){ animation-delay:.30s; }
@keyframes bob { 40%{ transform: translateY(-4px);} }
```

Don't animate per token. Fading in each streamed chunk thrashes layout. Animate at the **message** level (one fade-in on mount) or not at all; let text just appear.

Auto-scroll

- Stick to bottom with `scrollTo({top, behavior:"smooth"})` on each chunk...
- ...but **only if the user is already near the bottom**. If they scrolled up to read, stop auto-scrolling (check `scrollHeight - scrollTop - clientHeight`).

Animatable? — quick reference

Want to animate...	Do this instead
position (top/left)	<code>transform: translate</code>
size (width/height)	<code>transform: scale</code> (mind distortion)
height: auto	grid <code>0fr-1fr</code> / max-height / JS
display none ↔ block	<code>@starting-style+allow-discrete</code> , or visibility+opacity
show/hide	<code>opacity</code> (+ visibility)
box-shadow	opacity of a shadow pseudo-element
gradient / color	animate <code>opacity</code> of a layered element

How to talk about it in the loop

- Name it as a **step**: "I'll get behavior + states right, then add the slide transition and reduced-motion."
- When you pick `transform`, say **why** (compositor, no reflow). That one sentence is the signal.
- Call the trap before you hit it: "height auto won't transition, so I'll use the grid trick."
- Close with reduced-motion — reads as production maturity, not interview theater.